

Digital forensic analysis of base-wkstn-01-c-drive.E01

BASE-WKSTN-01 · EVID-001 · acquired 2021-09-16

CLIENT	shieldbase.lan / Blind triage
CASE ID	CASE-002 (Test3)
PREPARED BY	External DFIR analyst (SIFT Workstation)
REPORT DATE	2026-06-04 UTC
CLASSIFICATION	CONFIDENTIAL — RESTRICTED DISTRIBUTION

■ CONFIDENTIAL

Report generated 2026-06-04

01 Executive Summary

A single, **Confirmed** instance of commercial malware — **BlazingTools “Perfect Keylogger”** — was identified on the workstation, persistently installed in `C:\ProgramData\perfmon-k\`. The keylogger components were dropped on **2018-08-31 21:39:35 UTC** with **anti-forensic timestomping** that backdates the `$STANDARD_INFORMATION` timestamps by exactly one year (to 2017-08-31). The deployment was carried out by the **shieldbase\spsql** domain account through a Microsoft Edge download (`k.exe` — 14 MB self-extracting installer) followed by execution of an embedded WinRAR SFX (`setup.exe`) in the user’s Temp tree.

The `spsql` user profile has been **removed from the workstation** (no longer in the registry `ProfileList` and no `Users\spsql` directory), but a leftover scheduled-task file `OneDrive Standalone Update Task-S-1-5-21-3445421715-2530590580-3149308974-1193` preserves the deleted user’s SID and confirms the account existed.

McAfee VirusScan Enterprise 8.8 was installed on the host but **did not detect the keylogger**: `McShield.evtx` is empty (0 records) and `C:\Quarantine\` is empty.

Beginning **2020-02-22**, the **shieldbase** Incident Response team — chiefly user `cbarton-a` from threat-hunting workstations `BASE-HUNT-02` and `BASE-HUNT-03` — performed repeated authenticated F-Response Subject + Mnemosyne memory/disk acquisitions against this host. This was preceded on 2020-02-17 by a Group Policy Restricted-Groups change that added a domain account to the local `Builtin\Administrators` group, consistent with provisioning IR access.

11

KEYLOGGER COMPONENTS ON DISK

3 yr

DAYS RESIDENT (AV-EVADED)

11 / 11

FILES TIMESTOMPED (SI<FN)

156

LOGGED FORENSIC COMMANDS

02 Evidence Examined

PROPERTY	VALUE
Evidence ID	EVID-001
Source path	/cases/Test1/base-wkstn-01-c-drive.E01
Format	E01 (single NTFS volume — no MBR / partition table)
Size	33,833,353,216 bytes (31 GiB)
Acquisition tool	FTK Imager ADI4.2.0.13
Acquired	2021-09-16 03:10:21 UTC
MD5 (EWF metadata)	fd463e4eb6b5744c84b5ad138e7fec7f
SHA-1 (EWF metadata)	c4e37e7fe2d95db50d3c5a8687a5cf6aad101503

PROPERTY	VALUE
Verification	EWFVERIFY SUCCESS — hashes match acquisition
Mount	/mnt/ewf_wkstn01/ewf1 (fuse, ro) → /mnt/wkstn01 (NTFS, ro)
NTFS volume serial	82424B85424B7CC3

03

System Profile

PROPERTY	VALUE
Hostname	BASE-WKSTN-01
FQDN	base-wkstn-01.shieldbase.lan
Operating system	Windows 10 x64 (Build 16299 – Fall Creators Update 1709)
Domain	shieldbase.lan (SID prefix S-1-5-21-3445421715-2530590580-3149308974-)
Anti-virus	McAfee VirusScan Enterprise 8.8.011000 (installed 2018-05-10)
EDR / telemetry	Sysmon 64 (2019-12-24), Velociraptor service (2020-02-27)
Event forwarding	Windows Event Forwarding to remote collector (~210k subscription-enumeration events — benign)
Primary interactive user	shieldbase\mhill (Type-11 cached / Type-7 unlock; last 2020-09-21)

Profiles present (RID ↔ name):

RID	USERNAME	PROFILE PATH
500	administrator.shieldbase	C:\Users\administrator.shieldbase
1118	mhill	C:\Users\mhill
1183	rsydow-a	C:\Users\rsydow-a
1185	cbarton-a	C:\Users\cbarton-a
1193	spsql PROFILE DELETED	only OneDrive scheduled-task file preserves SID
1212	(identity unknown — no profile on this host)	added to local Administrators 2020-02-17 (GPO)

04

Tools and Versions

TOOL	VERSION	PURPOSE
ewfinfo / ewfverify	libewf 20140816	E01 metadata, hash verification
The Sleuth Kit (fsstat , fls , icat)	system	Volume metadata, MFT & \$UsnJrn1:\$J extraction
EZ Tools EvtxECmd	1.x	Event log → CSV (297 channels, 1.3 GB EVTX)
EZ Tools AmcacheParser	1.5.2.0	Amcache.hve → CSV (104 program entries, 2,111 unassociated files)

TOOL	VERSION	PURPOSE
EZ Tools AppCompatCacheParser	1.5.1.0	Shimcache → CSV (489 entries)
EZ Tools RECmd	2.1.0.0	Registry hive batch parsing (Kroll batch)
EZ Tools MFTECmd	current	\$MFT (276 MB) and \$UsnJrn1:\$J (9.0 GB) → CSV
YARA (python module)	4.5.0	Custom BlazingTools signature sweep
WeasyPrint	current	Report rendering

05

Findings — F-01 to F-09

F-01 — BlazingTools “Perfect Keylogger” in C:\ProgramData\perfmon-k\

▲ HIGH SEVERITY — Confirmed Commercial Spyware

Confidence: **Confirmed**. Family marker `BlazingTools Perfect Keylogger` is present in plaintext (UTF-16LE) inside `install.log`. The 11-file component layout exactly matches the published PKL component set.

On-Disk Component Table

FILE PATH	SIZE B	SHA-256 (TRUNCATED)	NOTES
C:\ProgramData\perfmon-k\install.log	520	4b641356... 7f6b945f	contains "BlazingTools Perfect Keylogger" plaintext
C:\ProgramData\perfmon-k\perfmon-kr.exe	1,352,704	fc7f5094... a4c19c75	Recorder executable
C:\ProgramData\perfmon-k\perfmon-kvw.exe	1,570,816	128cdb7c... a712185	Viewer / admin console
C:\ProgramData\perfmon-k\perfmon-khk.dll	1,307,648	225efff9... 5d3356c	Keyboard-hook DLL
C:\ProgramData\perfmon-k\perfmon-ki.dll	275,968	fa76d905... b4ad710	Injection helper DLL
C:\ProgramData\perfmon-k\perfmon-kwb.dll	1,348,608	8db5e3bd... a09407c	Browser-hook DLL
C:\ProgramData\perfmon-k\pk1.bin	9,368	5f7925d5... 142a78b	Collected-data database (PKL native format)
C:\ProgramData\perfmon-k\opt.dat	64	ebfafcb2... a9749ecd	Options binary
C:\ProgramData\perfmon-k\web.dt	304	8fe609ee... a3e07b59	Transport / upload config (potential C2)
C:\ProgramData\perfmon-k\install.bin	83,456	01907915... c06ebcd1	Installer payload binary
C:\ProgramData\perfmon-k\help.chm	407,279	beb9c597... 561e7b14c	Bundled help file

Timestamp Evidence (SI vs FN)

Every file in the keylogger directory has `$STANDARD_INFORMATION` timestamps stamped to **2017-08-31** but `$FILE_NAME` timestamps showing real creation on **2018-08-31**. MFTECmd flags `SI<FN = True` for all rows. Selected MFT entries (UTC):

FILE	SI CREATED (\$I0)	FN CREATED (\$30) — REAL	MFT ENTRY
<code>help.chm</code>	2017-08-31 17:30:35	2018-08-31 21:39:35	148033
<code>install.bin</code>	2017-08-31 17:30:35	2018-08-31 21:39:35	148034
<code>install.log</code>	2017-08-31 17:30:50	2018-08-31 21:39:50	148050
<code>perfmon-khk.dll</code>	2017-08-31 17:30:35	2018-08-31 21:39:35	148036
<code>perfmon-ki.dll</code>	2017-08-31 17:30:35	2018-08-31 21:39:35	148042
<code>perfmon-kr.exe</code>	2017-08-31 17:30:35	2018-08-31 21:39:35	148043
<code>perfmon-kvw.exe</code>	2017-08-31 17:30:35	2018-08-31 21:39:35	148040
<code>perfmon-kwb.dll</code>	2017-08-31 17:30:35	2018-08-31 21:39:35	148041
<code>pk1.bin</code>	2017-08-31 17:39:19	2018-08-31 21:42:43	3134
<code>web.dt</code>	2017-08-31 17:33:47	2018-08-31 21:42:47	120226
<code>opt.dat</code>	2017-08-31 17:39:03	2018-08-31 21:48:03	120972

The exact one-year date offset and consistent hour-of-day delta indicate the operator used a SI-timestamp tool such as SetMACE, timestamp.exe, or an equivalent PowerShell routine. The FN timestamps preserve ground truth because they cannot be modified via standard Windows APIs.

YARA confirmation: rule `BlazingTools_Perfect_Keylogger_Install_Log` matched `install.log` (`/cases/Test1/exports/yara_hits.csv`).

Alternative explanations considered. A legitimate “PerfMon K-edition” performance tool — **REJECTED** — the plaintext BlazingTools marker and exact PKL component naming convention rule this out. IR-team-planted analysis sample — not supported because the files predate the IR-collection campaign by 18 months and use an attacker-style drop path (Edge download → user Temp → ProgramData).

Persistence: Hypothesis. No keylogger entry was found in `HKLM\Software\Microsoft\Windows\CurrentVersion\Run`, the corresponding `WOW6432Node Run` key, the `mh11 HKCU Run` key, `HKLM\SYSTEM\CurrentControlSet\Services, Microsoft-Windows-TaskScheduler/Operational`, or `C:\Windows\System32\Tasks\`. Persistence mechanism unresolved — possible Active Setup, COM hijack, Applnit_DLLs, WMI subscription, or operator-on-demand execution. Validation step: dump all ASEP locations with Autoruns / RECmd recursive search and review `root\subscription WMI` namespace.

F-02 — Deployment chain: shieldbase\spsql via Microsoft Edge

Confidence: **Corroborated**. Source: Amcache + OneDrive scheduled-task file + MFT FN timestamps.

AMCACHE KEY LWT (UTC)	FILE	PATH
2018-08-31 21:14:30	onedrivesetup.exe	C:\Users\spsql\AppData\Local\Microsoft\OneDrive\update\onedrivesetup.exe
2018-08-31 21:38:29	setup.exe	C:\Users\spsql\AppData\Local\Temp\rarsfx0\setup.exe
2018-08-31 21:39:54	k.exe	C: \Users\spsql\AppData\Local\Packages\Microsoft.MicrosoftEdge_8wekyb3d8bbwe\T
2018-08-31 21:43:21	perfmon-k.exe	C:\ProgramData\perfmon-k\perfmon-k.exe (no longer present)
2018-08-31 21:48:03	perfmon- kconfigure.exe	C:\ProgramData\perfmon-k\perfmon-kconfigure.exe (no longer present)

The `spsql` user profile has been removed from this host. The only surviving SID reference is the file `c:\Windows\System32\Tasks\OneDrive Standalone Update Task-S-1-5-21-3445421715-2530590580-3149308974-1193` dated 2018-08-31, which preserves the deleted user's SID-to-presence binding.

F-03 — Anti-forensic tradecraft

Confidence: **Confirmed**. Three concurrent behaviours: (a) one-year SI timestamp on all keylogger files (see F-01), (b) `spsql` user-profile removal, eliminating NTUSER.DAT / browser history / ShellBags / JumpLists / RecentDocs, and (c) a backdated PE LinkDate on the downloaded `k.exe` (reports 2010-02-10).

F-04 — Local-Administrators escalation (likely IR provisioning)

Confidence: **Probable** (legitimate). One EID-4732 ("Member added to security-enabled local group") fired in the entire ~3.5-year log:

UTC	SUBJECT	GROUP	ADDED MEMBER SID
2020-02-17 08:00:37	shieldbase\BASE-WKSTN-01\$ (SYSTEM, 0x3E7)	Builtin\Administrators	S-1-5-21-3445421715-2530590580-3149308974-1212

The computer-account / SYSTEM subject and the absence of any interactive logon by that SID is consistent with GPO Restricted-Groups. Five days later `cbarton-a` began deploying F-Response (F-05). Identity of SID...-1212 requires AD lookup. Alternative explanation: attacker GPO write — weaker because only one such event exists in the log and a documented IR-collection campaign followed shortly after.

F-05 — Documented IR collection (2020-02 to 2021-09)

Confidence: **Confirmed**. Service installs (Security EID 4697, System EID 7045):

FIRST SEEN (UTC)	SERVICE	IMAGEPATH	INSTALLER SUBJECT
2019-11-13 02:53:48	mnemosyne	C:\windows\Mnemosyne.sys (F-Response kernel)	SYSTEM
2019-12-24 03:43:30	Sysmon64 / SysmonDrv	C:\WINDOWS\Sysmon64.exe	SYSTEM
2020-02-22 23:16:16	F-Response Subject	C:\windows\subject_srv.exe -s \	shieldbase\cbarton-a
2020-02-27 20:04:23	Velociraptor Service	(per System EID 7045)	SYSTEM

F-Response Subject was reinstalled ≥ 4 times by cbarton-a; mnemosyne driver service was started 19 times between 2018-09 and 2021-09. The image was acquired **9 minutes** after the final F-Response Subject service start (2021-09-16 03:01:57 → 03:10:21 UTC), i.e. captured remotely over F-Response.

Inbound SMB activity (Security EID 5140) supporting the IR campaign:

HITS	SHARE	USER	FROM IP
14,250	*\IPC\$	ANONYMOUS LOGON	172.16.7.13
2,641	*\IPC\$	ANONYMOUS LOGON	172.16.7.14
65	*\IPC\$	cbarton-a	172.16.5.25
15	*\IPC\$	rsydow-a	172.16.4.4
8	*\C\$	rsydow-a	172.16.4.4
8	*\TempFRes	cbarton-a	172.16.5.25
7	*\C\$	BASE-HUNT-03\$	172.16.5.28
4	*\ADMIN\$	cbarton-a	172.16.5.25
2	*\IPC\$	spadertech\pman.adm	172.16.6.17
1	*\ADMIN\$	BASE-HUNT-02\$	172.16.5.27

The TempFRes share is created by F-Response during forensic collection. The single cross-domain account spadertech\pman.adm warrants customer validation (2 IPC\$ hits only).

F-06 — Kansa PowerShell IR collection (2018-08-15)

Confidence: **Confirmed**. Microsoft-Windows-PowerShell/Operational contains 1,218 EID-4104 script-block events including Get-Tasklistv.ps1, Get-Procswmi.ps1 and the Kansa.ps1 output directive follows marker. Ran **16 days before** the keylogger install — not indicative of active compromise at that point.

F-07 — High-volume WinRM is Windows Event Forwarding (benign)

Confidence: **Confirmed**. 210,037 EID-145 events of which 209,791 enumerate the SubscriptionManager/Subscription resource URI on an 85-second cadence — classic WEF heartbeat.

F-08 — AV bypass

Confidence: **Confirmed**. McShield.evtx contains 0 records; C:\Quarantine\ is empty; no Application.evtx events name the BlazingTools components. The keylogger evaded McAfee for the full 3+ years.

F-09 — Local range_admin account

Confidence: **Probable** (cyber-range baseline artefact). SAM RID 1003 in Administrators group; password set 2018-08-29 (two days before the spsql incident); **0 interactive logons** in 3.5 years. Worth disabling on rebuild.

● 2018-05-04 18:14	Earliest event — Windows 10 install / domain join
● 2018-05-10 → 2018-05-14	McAfee VirusScan / HIPS / Agent services installed
● 2018-08-15 ~16:30	Kansa IR PowerShell collection executed (1,218 4104 script-blocks)
● 2018-08-29 02:57	range_admin local-account password set
● 2018-08-31 21:14	spsql OneDrive first-run (profile materialised)
● 2018-08-31 21:38:29	setup.exe (WinRAR SFX) executed in %TEMP%\rarsfx0\
● 2018-08-31 21:39:35	BlazingTools PKL files dropped to C:\ProgramData\perfmon-k\ (FN; SI timestamped)
● 2018-08-31 21:39:54	k.exe (14 MB BlazingTools installer) recorded in Amcache from Edge downloads
● 2018-08-31 21:42:43	pk1.bin (PKL data DB) created
● 2018-08-31 21:48:03	perfmon-kconfigure.exe last activity; install complete
● 2019-11-13 02:53	mnemosyne kernel driver service installed
● 2019-12-24 03:43	Sysmon64 deployed
● 2020-02-17 08:00:37	SID ...-1212 added to Builtin\Administrators via GPO
● 2020-02-22 23:16	First F-Response Subject install by cbarton-a — IR collection begins
● 2020-02-27 20:04	Velociraptor Service installed
● 2020-09-21 00:08	mhill last interactive logon
● 2021-09-16 02:52 → 03:01	Final mnemosyne / F-Response Subject demand-start
● 2021-09-16 03:10:21	E01 acquisition complete (FTK Imager)

TYPE	VALUE	CONFIDENCE
Folder	C:\ProgramData\perfmon-k\	HIGH
File path	%LOCALAPPDATA%\Packages\Microsoft.MicrosoftEdge_8wekyb3d8bbwe\TempState\Downloads\k.exe	HIGH
File path	%LOCALAPPDATA%\Temp\rarsfx0\setup.exe	HIGH
File names	perfmon-kr.exe , perfmon-kvw.exe , perfmon-khk.dll , perfmon-ki.dll , perfmon-kwb.dll , pk1.bin , opt.dat , web.dt	HIGH
String	BlazingTools Perfect Keylogger (ASCII / UTF-16LE)	HIGH
SHA-256 perfmon-kr.exe	fc7f5094bc182a05e782e0b775eccab9e662a3d207e89ab7f075db18a4c19c75	HIGH
SHA-256 perfmon-kvw.exe	128cdb7c58bfda11a88c21e41e48df47f2122a1761d69aec5af79e241a712185	HIGH
SHA-256 perfmon-khk.dll	225efff946151edeb6f860f5e4d68aa7025ca7abef67ddb215f5c89445d3356c	HIGH
SHA-256 perfmon-ki.dll	fa76d9059fc4fe103e3de1868c76d86efbac9295fcd3e5f1c430cd1c5b4ad710	HIGH
SHA-256 perfmon-kwb.dll	8db5e3bdd57ed77f80e1dbd5f11a48a19f69faa2281f1039b615ee8b4a09407c	HIGH
SHA-256 pk1.bin	5f7925d5e11232a4391daabd56559ad79e662d03e9d6f791b4ace3ec3142a78b	HIGH
SHA-256 install.log	4b64135662977d806f8d12067815848827cd977c3696a60c30d85a157f6b945f	HIGH
SHA-256 web.dt	8fe609eeaaea64bd6f3b5dd786be42d6ce46c41f5f7145a65932e0cab3e07b59	HIGH
SHA-1 setup.exe	40b638c05767cc5ad9802701953efeb5315bc80f	HIGH
SHA-1 k.exe	3bc3eb9c78d0f867f7d138f7958c2b8854a4dc53	HIGH
SHA-1 perfmon-k.exe (deleted)	c31bbf518e9d9bc74a6567a2e3f0c37043ad399a	HIGH
SHA-1 perfmon-kconfigure.exe (deleted)	c95f567910b56d5e1f865781582aa6c8ff9f5db9	HIGH
Username	shie1dbase\spsql (SID ...-1193, profile deleted)	HIGH
Behaviour	\$STANDARD_INFORMATION one calendar year earlier than \$FILE_NAME (SI<FN True)	MEDIUM

TYPE	VALUE	CONFIDENCE
YARA rule	BlazingTools_Perfect_Keylogger_Install_Log , BlazingTools_PKL_Filenames	HIGH
Source IP	172.16.6.17 (spadertech\pman.adm) — cross-domain IPC\$	MEDIUM

Full register: /cases/Test1/analysis/ioc_register.csv

08 Gaps and Limitations

What is known. All 11 BlazingTools PKL components are intact and hashed. Install on 2018-08-31 21:39:35 UTC by deleted domain user shieldbase\spsql via a Microsoft Edge download. Anti-forensic timestamping applied to all components; FN timestamps preserve ground truth. IR collection campaign by shieldbase security team active since at least 2020-02-22.

What is unknown.

- The persistence mechanism — none of the standard ASEP locations reference the keylogger.
- Transport / exfiltration channel — web.dt not decoded.
- Identity behind the spsql account (requires AD logs).
- Identity behind SID ...-1212 added to Local Administrators 2020-02-17.
- Whether keylogged data was exfiltrated (pk1.bin not decoded; no network capture).
- Whether other hosts in shieldbase.lan are similarly affected.

Evidence that would resolve uncertainty: AD logs for shieldbase.lan 2018-08, carved Edge WebCacheV01.dat fragments for spsql, decoded pk1.bin and web.dt, McAfee ePO logs, memory captures from F-Response sessions.

09 Recommendations

Immediate

1. **Pivot to other shieldbase.lan hosts** with the YARA rule at /cases/Test1/analysis/yara/blazingtools_pkl.yar — look for C:\ProgramData\perfmon-k* and the SHA-256 IOC set.
2. **Audit shieldbase AD** for the spsql account history and SID ...-1212 identity.
3. **Decode pk1.bin and web.dt** to recover captured keystrokes and the operator's exfiltration destination(s).

Follow-on

4. **Carve unallocated space** for Edge WebCacheV01.dat fragments and BAM/DAM database entries referencing spsql (SID ...-1193).
5. **Add custom McAfee signatures** for the BlazingTools SHA-256 set.
6. **Disable local range_admin** account on production rebuilds.
7. **Re-image this workstation;** do not return to service until persistence mechanism is confirmed and removed.

ARTEFACT	PATH
Command ledger	/cases/Test1/logs/command_ledger.csv
Evidence inventory	/cases/Test1/analysis/evidence_inventory.csv
Mounts register	/cases/Test1/analysis/mounts.csv
IOC register	/cases/Test1/analysis/ioc_register.csv
YARA rules	/cases/Test1/analysis/yara/blazingtools_pk1.yar
YARA hits	/cases/Test1/exports/yara_hits.csv
Parsed Amcache	/cases/Test1/analysis/amcache_*.csv
Parsed Shimcache	/cases/Test1/analysis/appcompatcache.csv
Parsed MFT	/cases/Test1/analysis/mft.csv
Parsed UsnJrnl	/cases/Test1/analysis/usnjrnl.csv
Parsed event logs (all)	/cases/Test1/analysis/evtx_all.csv
Parsed Security log	/cases/Test1/analysis/evtx_security.csv
Parsed System log	/cases/Test1/analysis/evtx_system.csv
Parsed Application log	/cases/Test1/analysis/evtx_application.csv
RECcmd Kroll batch	/cases/Test1/analysis/recmd/20260604160422/
Markdown report	/cases/Test1/reports/base-wkstn-01-c-drive-report.md
This PDF	/cases/Test1/reports/base-wkstn-01-c-drive-report.pdf

Command Ledger Summary

METRIC	VALUE
Total commands logged	156
Failed commands	31 (all investigative dead-ends: mmIs on partition-less image, empty greps, deprecated PECmd path, expected find -iname misses)
Rerun commands	1
Unresolved command failures	0
Audit script	/home/sansforensics/.claude/scripts/audit_command_ledger.py
Audit verdict	PASSED — ledger has correct headers, all stdout/stderr paths present, every row carries the seven required fields.

CONFIDENCE	DEFINITION
Confirmed	Directly supported by raw or parsed forensic artefact.
Corroborated	Supported by two or more independent artefacts.
Probable	Supported by one artefact plus strong contextual evidence.
Hypothesis	Plausible but not yet proven; must include the next validation step.
Rejected	Tested and unsupported.

This report was produced as part of an active digital forensics investigation. All findings are based on evidence present at the time of analysis. Evidence integrity maintained per chain-of-custody protocol — source images not modified.